

EMPLOYEE ATTRITION ANALYSIS

Reducing Attrition from 16.1% to Below 10%

People Analytics Division | IBM HR Analytics Dataset

Prepared for CEO & CHRO

The organization is losing 1 in 6 employees — costing \$4M–\$7M+ annually

THE PROBLEM

Attrition rate: 16.1%

Target: < 10%

Replacement cost:

50–200% of annual salary
per employee (SHRM)

KEY DRIVERS

1. Overtime → 3x attrition
2. Early career → 44% leave
3. Low salary + OT → 58.5%
4. No stock options → 24.4%

RECOMMENDATION

4 targeted interventions

Invest: \$520K–\$750K +
equity

Save: \$4M–\$7M+

ROI: 5–10x return

At 16.1%, attrition is above industry benchmark — each departure costs 50–200% of annual salary

16.1%

Current
Attrition Rate

237

Employees
Left

≤ 10%

Industry
Best Practice

90+

Departures
to Prevent

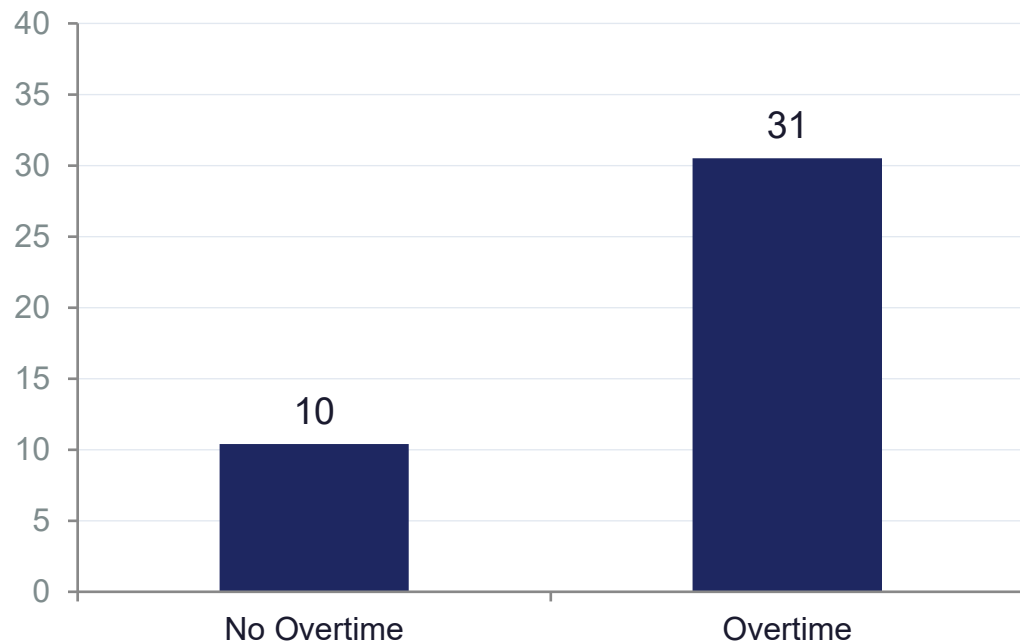
COST IMPACT

Replacing one employee costs 50–200% of their annual salary (SHRM). At 237 departures, the organization faces \$4M–\$7M+ in annual replacement costs.

A full end-to-end analytics workflow — from problem to action plan



Employees who work overtime are 3x more likely to leave

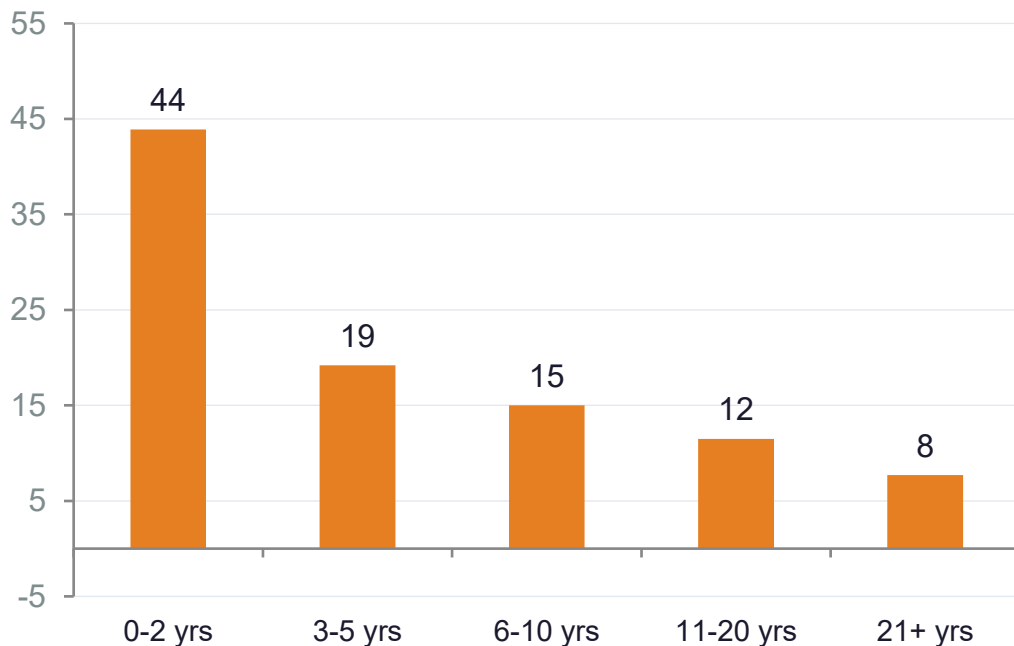


30.5%

of overtime employees leave
vs 10.4% without overtime

416 employees (28.3%)
work overtime

Early-career employees leave at 3–4x the rate of experienced staff



43.9%

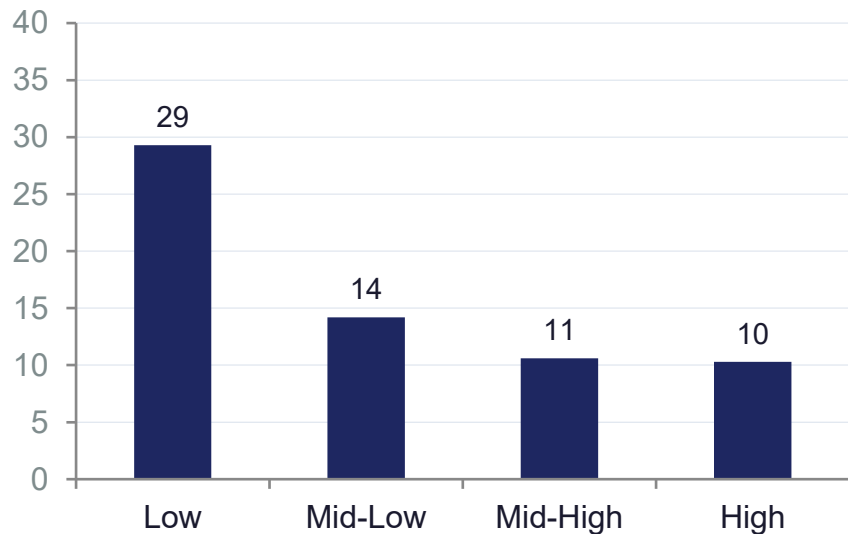
of 0–2 year employees leave
Nearly half walk out the door

First-year tenure: 34.9%

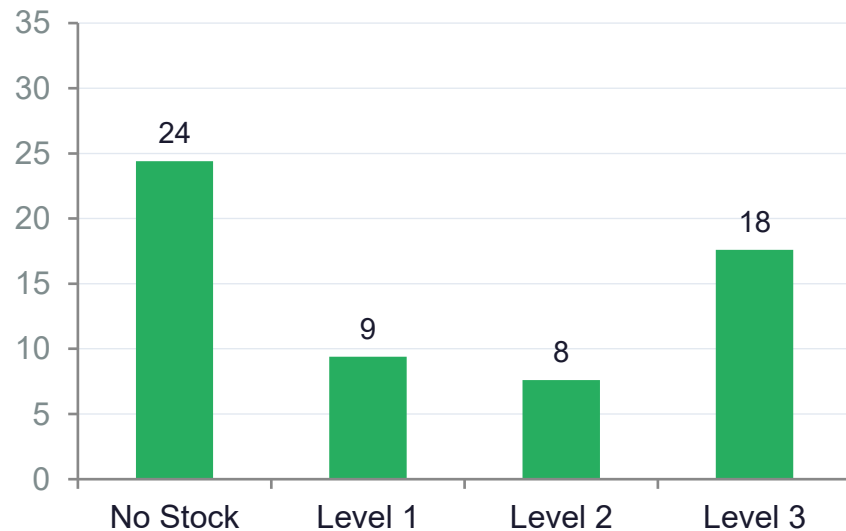
Job Level 1: 26.3%

Low salary drives 3x higher attrition — stock options cut risk by 32% per level

Attrition by Salary Quartile



Attrition by Stock Option Level



Employees who leave earn 38% less (median \$3,202 vs \$5,204). Even Level 1 stock options reduce attrition from 24.4% to 9.4%.

When overtime meets low salary, more than half leave

58.5%

attrition rate for employees with BOTH overtime AND low salary

Low Salary only: 17.5%

Overall average: 16.1%

106 employees in this group — the highest-priority intervention target

We built a predictive model that catches 62% of at-risk employees before they resign

MODEL PERFORMANCE

62%

Recall (catch rate)

Catches 3 in 5 at-risk employees

0.74

ROC-AUC

74% better than random

0.48

PR-AUC

Best among all models tested

WHY THIS MODEL?



Interpretable — explains WHY, not just who



Actionable — each factor maps to an intervention



Validated — outperforms RF on recall & PR-AUC



Cost-justified — miss = \$39K+, false alarm = a chat

181 employees (12.3%) are High Risk — with 51.9% actual attrition rate

Risk Category	Employees	% Workforce	Actual Attrition	Action
HIGH RISK	181	12.3%	51.9%	Immediate intervention
MEDIUM RISK	487	33.1%	18.1%	Proactive monitoring
LOW RISK	802	54.6%	6.9%	Standard engagement

76.8%

of all attrition captured in High + Medium risk groups.
HR focuses on 668 employees (45%) instead of all 1,470.

The typical high-risk employee: overtime, low pay, early career, no stock options



OVERTIME

77%

work overtime



LOW INCOME

\$2,994

avg monthly
(vs \$6,503 avg)



EARLY CAREER

61%

0-2 years
experience



NO STOCK

67%

no stock
options

Not a vague label — a clear, actionable profile HR can identify and act on immediately.

4 targeted interventions — prioritized by impact per dollar spent

P1	P2	P3	P4
Overtime Reduction	Stock Option Grant	Early-Career Retention	Compensation Review
416 emp	631 emp	242 emp	138 emp
Cost: Low	Cost: Medium	Cost: Low	Cost: TBD
★★★★★	★★★★★	★★★★★	★★★

P4: Benchmark against market first, then decide. Do not assume pay is below market without data.

Invest \$520K–\$750K to save \$4M–\$7M+ in replacement costs

INVESTMENT

\$520K–\$750K

+ equity allocation

Overtime reduction (process)

Stock option grants (equity)

Onboarding program (~\$50K)

Partial salary adjustment



RETURN

\$4M–\$7M+

in avoided replacement costs

90+ departures prevented

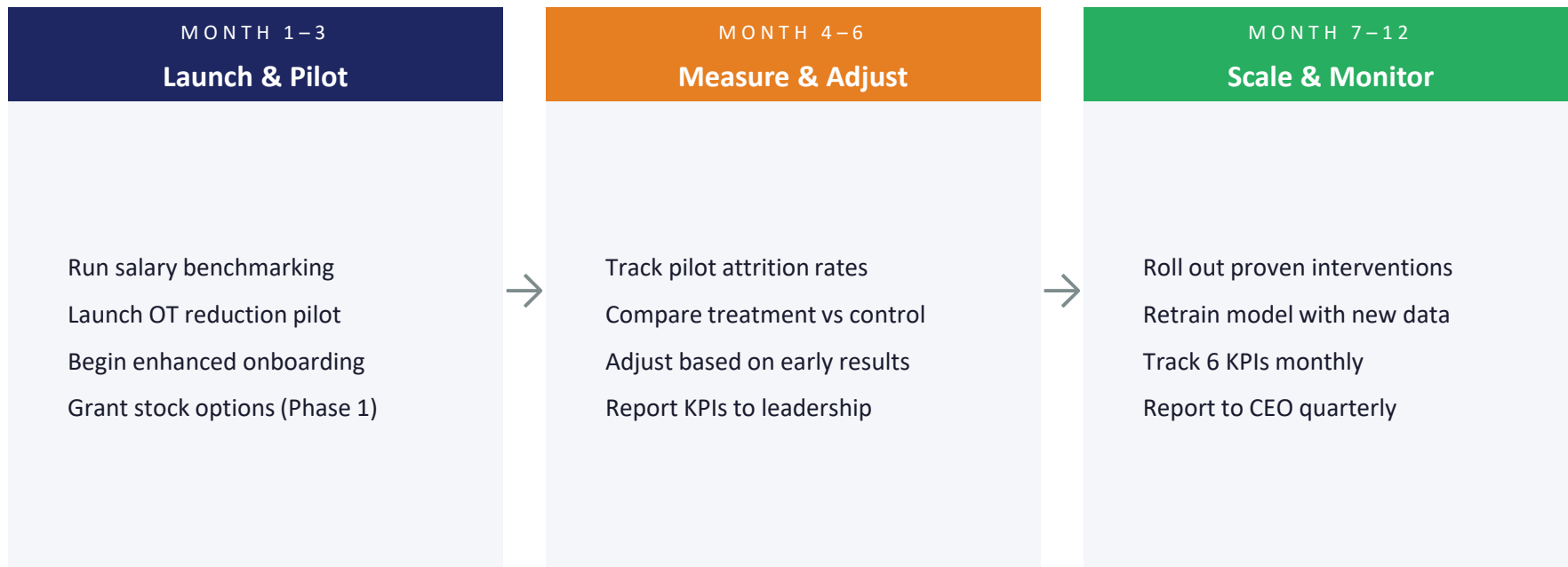
Attrition: 16.1% → below 10%

ROI: 5–10x return

Break-even in < 6 months

The interventions pay for themselves many times over.

Pilot in 3 months, measure in 6, scale by month 12



We request approval to proceed with the pilot phase starting next month.